

YASHITA TANEJA

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Education

Sophomore at University of California, San Diego

B.S. in Mathematics-Computer Science, Minor in Data Science

Skills

Languages: Python, Java, C, C++

Tools: NumPy, pandas, PyTorch, statsmodels, pgmpy, MySQL, Git, networkx

Concepts: Data Structures & Algorithms, Data Analysis, Machine Learning, Statistical Modeling

Experience

Data Analyst Intern, Exotic India

May 2023-June 2023

- Collected, analyzed, and categorized sales data
- Actively involved in data mining and drawing conclusions and trends

Projects

Heatwave Prediction using Bayesian Networks:

Jan 2026-March 2026

- Built a heatwave prediction model using Bayesian Networks on 91 thousand+ daily climate records from India, leveraging temperature, precipitation, wind, season, and city-level features.
- Engineered lagged and threshold-based heatwave features, applied time-based evaluation, and achieved 1.00 precision, 0.95 recall, and ~0.98 balanced accuracy in the final model.

Representation Manipulation via Representation Reading:

Jan 2026-March 2026

- Replicated an LLM interpretability study by using TinyLlama-1.1B to analyze how hidden activations encode causal v/s correlational reasoning.
- Generated 400 synthetic prompt pairs, extracted a causal reading vector using PyTorch hooks and paired-difference PCA, demonstrated inference-time activation and achieved 98.2% accuracy on synthetic validation tasks.

Socioeconomic Factors and Heart Disease Mortality Analysis:

Sept 2026-Dec 2026

- Integrated and cleaned public health and economic datasets for 58 California counties to analyze links between socioeconomic disadvantage and heart disease mortality.
- Applied EDA, visualization, and multiple linear regression, showing that lower income and higher unemployment were associated with higher mortality, with the model explaining 44% of observed variation.

GYM Management System:

Sept 2023-Dec 2023

- Developed a web-based gym management system for trainer registration, member management, and equipment tracking using Python and MySQL.
- Built an admin panel to centralize operations and streamline record management across multiple functions.